

Locally Robust Commensurate BART for Borrowing External Controls

Model local compatibility and calibrate borrowing

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1. Established borrowing priors decide through one study-level parameter; whether an external patient is a fair control depends on who that patient is.
2. LRC-BART writes the RCT control surface as a pooled BART surface plus a sparse discrepancy ensemble, with a spike-and-slab commensurate prior on each discrepancy leaf.
3. A variance-based prior ESS is defined at the standardized control mean; its ceiling depends on the RWD fit at trial profiles.
4. Simulations with regional shifts, and a 30-patient single-arm myeloma trial with a registry control.

Potential gain

- More control observations
- Greater precision
- Smaller concurrent control arms

Why pooling can mislead

- Different eligibility and care pathways
- Different covariate distributions
- Measured and unmeasured confounding

External controls that match the trial well for one kind of patient can be badly off for another. A single borrowing parameter cannot tell them apart.

Where the established priors make the decision

Meta-analytic predictive prior (Neuenschwander et al., 2010; Schmidli et al., 2014)

Study parameters exchangeable,
 $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$; the robust version mixes in a vague component so that conflict switches borrowing off.

Commensurate prior (Hobbs et al., 2011, 2012)

$\theta \mid \theta_0 \sim N(\theta_0, 1/\tau)$ with a spike-and-slab hyperprior on the commensurability τ .

Power prior (Ibrahim and Chen, 2000)

Historical likelihood raised to a_0 .

Common feature

One heterogeneity or discount scale for the whole population, with no dependence on the patient's covariates. With a single historical study the three forms nearly coincide (Röver and Friede, 2025).

We make the borrowing decision region by region in covariate space, with the regions learned from the data.

The model in one line

RCT controls ($D_i = 1$) and RWD controls ($D_i = 2$), covariates $\mathbf{x}_i$, outcome y_i :

$$y_i = f(\mathbf{x}_i) + \mathbb{I}\{D_i = 1\} g(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_{D_i}^2)$$

f : external-control response mean

Standard BART, $H_f = 50$ trees, fitted to both sources. The RWD follow f .

g : discrepancy ensemble

Small and shallow, $H_g = 5$ trees, split prior $(0.5, 3)$, informed by the RCT controls only. The RCT controls follow $f_1 = f + g$.

- The decomposition is that of Bayesian causal forests (Hahn et al., 2020), with the source in the place of the treatment.
- The discrepancy is $g = f_1 - f$. Shrinking it toward zero centers the RCT-control mean f_1 at the external-control mean f .

The leaf prior of the discrepancy ensemble

The zero-centered mixture is on a leaf contribution to $g = f_1 - f$:

$\theta_{h\ell}$ is the contribution of leaf (h, ℓ) to g :

$$\theta_{h\ell} \mid w, \tau_0^2, \tau_1^2 \stackrel{\text{iid}}{\sim} w \text{N}(0, \tau_0^2) + (1 - w) \text{N}(0, \tau_1^2).$$

Equivalently, $z_{h\ell} \mid w \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w)$ selects a zero-mean normal with variance $\tau_{1-z_{h\ell}}^2$. We use $w \sim \text{Beta}(1, 1)$, $\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)$ truncated to $(0, \tau_1^2)$, and τ_1^2 fixed by the range rule.

Spike, $z = 1$: the commensurate component

Shrinks a discrepancy contribution toward zero, drawing the RCT response mean toward $f(x)$. The ESS calibration sets the leaf scale τ_0 .

Slab, $z = 0$: the vague component

A free local offset on the outcome scale: the robust-MAP vague component placed inside a leaf.

Special cases

$w = 1$: commensurate BART (C-BART), one learned scale. $w = 1, \tau_0^2 \rightarrow 0$: BART on the pooled controls (BART-CP). $w = 0$: a free source offset, the tree analogue of BART with the source indicator (BART-PP).

The induced prior is centered at $f(\mathbf{x})$

- At a fixed profile $\mathbf{x}$, each discrepancy tree contributes one leaf: $g(\mathbf{x}) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(\mathbf{x})}$, where $\ell_h(\mathbf{x})$ is the reached leaf.
- Integrating the leaf indicators gives a mixture over the number m of reached spike leaves:

$$f_1(\mathbf{x}) \mid f, T_g, w, \tau_0^2, \tau_1^2 \sim \sum_{m=0}^{H_g} \binom{H_g}{m} w^m (1-w)^{H_g-m} \mathcal{N}(f(\mathbf{x}), m\tau_0^2 + (H_g - m)\tau_1^2).$$

All spike: $m = H_g$

Variance $H_g \tau_0^2$: the commensurate component shrinks $f_1(\mathbf{x})$ toward $f(\mathbf{x})$.

All slab: $m = 0$

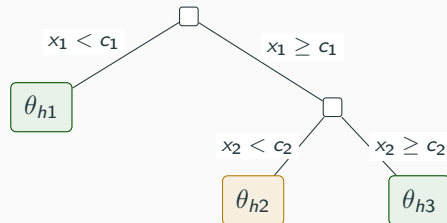
Variance $H_g \tau_1^2$: a larger departure is permitted, with the same center $f(\mathbf{x})$.

- T_g denotes the discrepancy partitions; every component is centered at the external mean $f(\mathbf{x})$.
- This is a prior for the latent control mean. Observation variance is separate.

A shared surface with local discrepancies

RWD controls: $f(x)$ **RCT controls:** $f(x) + g(x)$

One tree in g (illustrative splits)



Each profile reaches one leaf per tree.
Add those contributions: $g(x) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(x)}$.

f : the external response mean

Fitted jointly to both control sources.

Spike: $z_{h\ell} = 1$

$\theta_{h\ell} \sim N(0, \tau_0^2)$

Draws the RCT mean toward $f(x)$.

Slab: $z_{h\ell} = 0$

$\theta_{h\ell} \sim N(0, \tau_1^2)$, $\tau_1^2 > \tau_0^2$

Allows departures from $f(x)$.

$z_{h\ell} \sim \text{Bernoulli}(w)$. Leaf indicators vary;
 w, τ_0^2, τ_1^2 are shared.

Why a separate discrepancy ensemble

- Separate ensembles permit different partition and shrinkage priors for the external mean f and the discrepancy g .
- We use $H_f = 50$ trees for the outcome surface and $H_g = 5$ shallow trees for the discrepancy.
- At fixed leaf variances and partitions, fewer discrepancy trees reduce the total all-spike variance at a profile. The slab range rule also depends on H_g .

Scope of the prior-cost calculation

Appendix G.6 minimizes over integer numbers of slab leaves and retains contributions from the remaining spike leaves. It compares joint prior densities at constrained modes; posterior indicator probabilities also depend on the likelihood and partitions.

The simulations assess this specification. They do not establish that shared-leaf models cannot detect regional discrepancies.

Every leaf marginal is exact

Conditionally on (τ_0^2, τ_1^2, w) the leaf prior is a finite Gaussian mixture, so each tree move uses closed-form factors:

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp\left\{\frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)}\right\}, \quad M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)},$$

with A, B the precision-weighted counts and sums of both sources in an f leaf, and $\bar{y}_1, v_1 = \sigma_1^2/n_1$ the RCT residual mean and its variance in a g leaf.

- Grow, prune and change moves use the exact collapsed ratios, so the sampler targets the exact posterior: no surrogate likelihood.
- Leaf draws are conjugate; z is drawn first, then θ ; τ_0^2 and w update from the leaf means and states.
- Right-censored outcomes: log-normal AFT with data augmentation; nothing else changes.
- One sweep costs a standard BART on $n_1 + n_2$ observations plus a five-tree ensemble on n_1 .

Prior effective sample size at the estimand

For $\mu = N^{-1} \sum_i f_1(x_i)$, V_μ^f is the RWD-only posterior variance. The working calibration is:

$$\text{ESS}_0(s_0^2) = \hat{\sigma}_1^2 \mathbb{E}_{\tau_0^2} \left[\frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right].$$

- c_g averages squared leaf proportions before inversion; the scale law here is untruncated.
- A normal prior gives the ELIR variance ratio (Neuenschwander et al., 2020); this BART/scale-mixture summary is not exact marginal ELIR.
- Ceiling: $\hat{\sigma}_1^2 / V_\mu^f$, including BART regularization. Equality to scaled n_2 requires one fixed tree, matched proportions and a flat prior.
- The Pr-rule requires the target in 95% of blocks. A target at or above the whole-chain ceiling instead triggers a smallest-scale override.

Realized information diagnostic

With target leaf proportions $a_{h\ell}$, use $V_g = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell} \tau_0^2 + (1 - z_{h\ell}) \tau_1^2\}$, then average $\sigma_1^2 / (V_\mu^f + V_g)$. Standardization can span several leaves per tree.

Three results

Leaf results condition on f , the other discrepancy trees, partitions, scales and $0 < w < 1$. Write $v = \sigma_1^2/n$ for a leaf's RCT residual-mean variance.

Theorem 1: bounded local influence

Relative to the slab estimate, $\sup_{\bar{y}} |I(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}$, and the shift vanishes at a Gaussian rate under growing conflict. The C-BART and BART-CP leaf shifts are linear and unbounded.

Theorem 2: detection and the map

The leaf spike probability decreases with $|\bar{y}|$. With fixed partitions, f fixed, a representable discrepancy and growing local counts, $\mathcal{B}_c(x) \rightarrow \mathbb{I}\{|d| < c\}$ for $|d| \neq c$. A fixed positive spike scale need not identify compatibility through its indicator.

Theorem 3: the working calibration

Under the untruncated law, ESS_0 is continuous, convex and strictly decreasing from its ceiling to zero. Truncation gives a positive lower limit. Grid-selection consistency requires the stated moment and separation conditions; ceiling overrides are separate.

The borrowing map

For each patient profile x the analysis reports

$$\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}),$$

the posterior probability that the local discrepancy is below a threshold c fixed in advance on the outcome scale, together with the posterior mean and interval of $g(x)$ and the pointwise ESS.

- $c = 0.5$ in the simulations, one third of the RCT residual SD for Gaussian outcomes.
- The all-spike probability $B(x)$ is kept as a diagnostic: its prior scale is w^{H_g} and at a fixed positive spike scale, it need not identify compatibility as RCT counts grow.
- In single-arm mode nothing informs g , the map is flat by construction, and the covariate-resolved display is the local ESS map.

What is proved and what is not

The map results require fixed g partitions, a representable discrepancy, growing local counts and f fixed; there is no statement about the posterior over partitions. Separation of shifted from compatible regions under the full posterior is shown empirically.

Simulation design

Common design

- $R = 500$ paired replicates
- $p = 10$ covariates, nonlinear surface
- 300 RWD controls
- Gaussian: RCT 200:100, $\sigma_1 = 1.5$, $\sigma_2 = 1.8$
- Survival: RCT 225:75, log-normal AFT, about a quarter censored
- Targets: absolute 50, 75, 100; fractional 0.25, 0.5, 0.9 of the ceiling

Five scenarios

- **Sc1** no confounding: source at random
- **Sc2** measured: source depends on (X_5, X_6)
- **Sc3** unmeasured: source and outcome depend on $\mathbf{U}$, observed as proxies at $\rho \in \{-0.5, 0, 0.5\}$
- **Sc4** regional shift: RWD surface raised by $\delta_2 \in \{0.5, 1, 2\}$ outside $\mathcal{R} = \{X_5 > 2\}$, half the population; secondary version on the null covariate X_7
- **Sc5** global shift: raised by δ_2 everywhere; the right answer is to abstain

Illustrative leaf threshold: $d_{1/2} \approx 0.5$. Comparators: regression/BART pooling variants, PSCL, MAP, hierarchies, C-BART and LRC-BART ($w = 0$). CAHB (Jin et al., 2023): transferred analysis, first 200 paired Gaussian replicates; variants in the supplement.

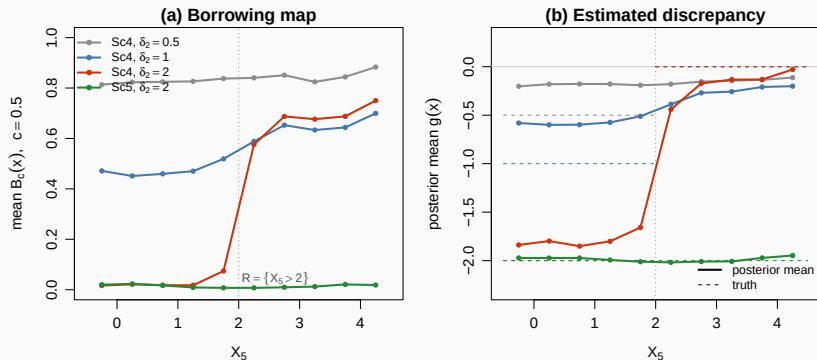
Gaussian outcomes: ATE bias, RMSE, coverage

Method	Sc1			Sc3, $\rho = 0$			Sc4, $\delta_2 = 1$			Sc4, $\delta_2 = 2$			Sc5, $\delta_2 = 2$		
	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov	Bias	RMSE	Cov
LM-PP	-0.00	0.34	0.94	0.04	0.35	0.95	-0.01	0.34	0.95	-0.02	0.34	0.95	-0.04	0.34	0.94
BART-NP	0.00	0.22	0.91	-0.00	0.29	0.94	0.00	0.22	0.91	0.00	0.22	0.91	0.00	0.22	0.91
BART-CP	0.01	0.14	0.96	0.98	1.00	0.00	-0.34	0.37	0.35	-0.68	0.70	0.00	-1.39	1.40	0.00
BART-PP	0.01	0.21	0.94	0.03	0.27	0.94	-0.00	0.21	0.95	-0.01	0.21	0.96	-0.01	0.21	0.95
C-BART	0.01	0.15	0.95	0.23	0.38	0.87	-0.21	0.27	0.73	-0.23	0.39	0.66	-0.32	0.65	0.76
LRC-BART (100)	0.01	0.17	0.94	0.04	0.28	0.94	-0.04	0.22	0.89	0.00	0.20	0.92	-0.00	0.20	0.92

Reading the table

Agreement (Sc1): LRC-BART borrows, RMSE 0.172 against 0.205 for BART-PP; the pooling models win here. Global disagreement (Sc3, Sc5): it abstains, bias about 0.04 and 0.00, where C-BART and BART-CP fail. Regional shift: near-zero empirical bias at $\delta_2 = 2$; at $\delta_2 = 1$ it borrows across the boundary, bias -0.04 , coverage 0.89.

The borrowing map under a regional shift



LRC-BART at target 100, first 40 replicates

At $\delta_2 = 2$ the map is 0.02 outside $\mathcal{R}$ and 0.7 inside, and g is -1.8 and -0.15 against -2 and 0 ; the transition occupies the unit around the boundary, learned from about four RCT controls per 0.1 of X_5 . At $\delta_2 = 1$ the estimate splits the shift ($-0.6 / -0.25$). Under Sc5 the map is 0.01 everywhere.

What the Gaussian study shows

Agreement and global conflict

Sc1: RMSE 0.172 versus 0.205 for BART-PP; realized ESS 53, prior ESS 82, ceiling 127. Under global conflict, realized ESS falls to 1.6 (Sc3) and 0.2 (Sc5). Sc3 bias is 0.012 above BART-PP (paired SE 0.001); the leaf theorem does not isolate its cause.

Regional conflict remains difficult

At shift 2 (1.3 residual SDs), compatible-region surface RMSE over all RCT profiles is 0.824 versus 0.785. On control profiles away from the boundary, it is 0.747 versus 0.723 (difference 0.024, SE 0.005, $R = 200$). At shift 1 (two thirds of a residual SD), bias is -0.04 , coverage 0.89, and RMSE 0.223 versus 0.207.

Compatibility and realized regional ESS are distinct

At shift 2, realized regional ESS is 9.5 in the compatible region and 0.44 outside ($R = 200$). Each uses its own standardized mean; they are not additive. The compatibility map reports a posterior probability.

The CAHB comparison assesses our transfer of kernel-local estimation; it does not establish a general method ranking. Sc2's ceiling falls from 127 to 23, so absolute targets are capped.

Method	Sc1, median ratio			Sc3 $\rho = 0$, median ratio			Sc3 $\rho = 0$, RMST ratio		
	Bias	RMSE	Pow	Bias	RMSE	Cov	Bias	RMSE	Cov
AFT-NP	0.18	0.67	0.12	0.23	0.83	0.94	-0.03	0.15	0.92
BART-NP	0.26	0.51	0.46	0.19	0.57	0.90	0.06	0.17	0.92
BART-CP	0.06	0.23	0.63	2.31	2.41	0.00	0.59	0.61	0.01
BART-PP	0.03	0.35	0.25	0.04	0.45	0.91	-0.01	0.14	0.92
C-BART	0.03	0.25	0.42	0.58	0.91	0.88	0.14	0.22	0.85
LRC-BART (100)	0.05	0.31	0.34	0.15	0.56	0.94	0.03	0.15	0.94

Two readings

With 75 controls and censoring the median ratio has a wide, skewed posterior whose posterior mean has upward bias in Sc1; borrowing reduces RMSE (Sc1: RMSE 0.312 against 0.345, power 0.34 against 0.25 for BART-PP). Under unmeasured confounding LRC-BART abstains (realized ESS 2.3) but its wider posterior, together with the nonlinear ratio transformation, accompanies 0.11 more median-ratio bias than BART-PP; on the RMST ratio the biases are 0.03 and -0.005.

The single-arm design

Trial contributes treated subjects only ($n_T = 200$ Gaussian; 30 or 200 survival); the 300 RWD controls are the sole control source; LRC-BART in single-arm mode with $w = 1$.

No confounding

BART-CP and LRC-BART have Gaussian RMSE 0.16; the survival RMST ratio at $n_T = 30$ has RMSE 0.16 to 0.19 with coverage 0.92 to 0.97.

Measured confounding

Every method is biased: Gaussian ATE bias 1.68 (LM-CP), 0.41 (BART-CP), 0.42 (LRC-BART); no internal control identifies a discrepancy, and RWD overlap is limited.

- The single-arm ceiling is 27 to 34 under Sc2, so an absolute target of 100 is capped and LRC-BART approximates pooling with $w = 1$; the fractional target still widens the interval (coverage 0.82 to 0.99).
- With $w < 1$, slab variance $(1 - w)H_g\tau_1^2$ adds uncertainty: SD 1.4 for the ATE and 10 to 34 for the median ratio. Zero-mean g preserves linear log-time means, but survival-ratio means also change. Primary analysis: $w = 1$.
- Hierarchical regressions: ATE SD 8 under the stated hyperprior.

EloKRd phase II trial

- $n = 30$, all received EloKRd
- 13 PFS events; 10 deaths
- 50% high-risk cytogenetics; 30% ASCT

UCMM registry control

- $n = 253$ after cohort restrictions
- Standard induction regimens
- Five baseline covariates plus ASCT after induction

Coding and interpretation

Incompatible non-Hispanic dummy columns change extrapolation. Harmonization raises the working ESS ceilings from 12.6/14.6 to 63.7/36.7 (PFS/OS). ASCT is recorded after induction, so this adjustment supports a descriptive comparison, not a total causal treatment effect.

Estimand: standardized 5-year RMST ratio (neither trial median is reached). Kaplan–Meier: 1.015 for PFS, 0.90 for OS.

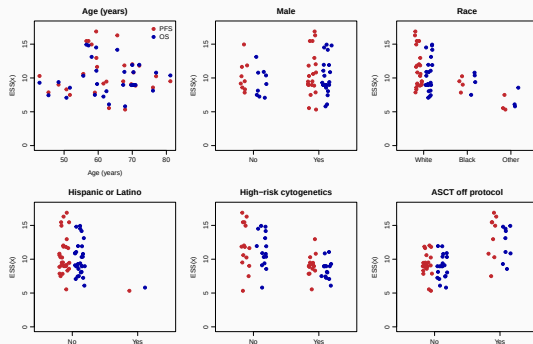
EloKRd results: 5-year RMST ratio, posterior mean and 95% interval

Harmonized coding	PFS	OS
Kaplan–Meier, unadjusted	1.01 (0.84, 1.22)	0.90 (0.79, 1.03)
Complete-pooling AFT	0.87 (0.69, 1.06)	0.80 (0.65, 0.94)
Hierarchical AFT	1.23 (0.74, 1.90)	1.21 (0.77, 1.70)
AFT-BART on UCMM, no discrepancy	0.97 (0.79, 1.16)	0.90 (0.78, 1.02)
LRC-BART, $N_{\text{target}} = 30$	0.97 (0.77, 1.23)	0.90 (0.77, 1.04)
LRC-BART, $N_{\text{target}} = 15$	0.99 (0.75, 1.40)	0.91 (0.76, 1.11)
LRC-BART, $0.9\times$ ceiling	0.97 (0.79, 1.15)	0.90 (0.77, 1.04)
<i>Earlier analysis, original coding: MAP-AFT-BART, 30</i>	1.08 (0.66, 16.4)	0.97 (0.74, 4.77)

Descriptive comparison conditional on recorded profiles

At target 30, $P(\text{ratio} > 1)$ is 0.37 (PFS) and 0.08 (OS); both nonparametric intervals include one. Lower targets widen intervals. The fitted PFS distribution has a 99.9th percentile of 1.6 over 8,000 draws, but the unavailable earlier sampler prevents a controlled attribution of the tail difference.

The local ESS map at the 30 trial profiles



Reading the map

$ESS(x) = \hat{\sigma}_1^2 / \{V_f(x) + H_g \bar{\tau}_0^2\}$ ranges from 5 to 17 for PFS and 6 to 15 for OS: highest for White/ASCT/standard-risk profiles; lowest for the one Hispanic patient, Other/Unknown race and high-risk profiles without ASCT. $\bar{\tau}_0^2$ is the fitted mean scale. Target 30 concerns the standardized log-time mean; local values are not additive.

1. The RCT control mean is centered on the external mean through a spike-and-slab discrepancy ensemble.
2. Closed-form leaf marginals support posterior-invariant tree updates; this does not establish finite-chain mixing.
3. Variance-based ESS calibration concerns the standardized control mean. Its ceiling and approximation convention are explicit.
4. Conditional leaf influence and map results require the stated fixed-surface and partition assumptions.
5. Gaussian Sc1 RMSE is about 16% lower than BART-PP's. Regional conflict produces informative compatibility summaries, but also surface losses and excessive borrowing at moderate shifts.

Limitations and next steps

- Full-posterior map consistency over learned partitions is unproved; the Sc4 surface loss persists away from the boundary.
- Calibration uses a working precision summary. Exact marginal ELIR would require a different calculation and new calibration analyses.
- Multiple external sources require joint calibration. Additive ceilings have not been established.
- A single-arm study cannot learn the control discrepancy. The application remains a descriptive comparison with post-induction ASCT adjustment.
- Confirmatory use requires operating-characteristic evaluation, including Type I error and power, and a pre-specified role for the map.

Reported summaries

The compatibility map, prior pointwise ESS and realized regional ESS answer different questions. None establishes an observed count of external patients used at each profile.

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